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 Studierichting: Informatica
 Oefeningenreeks 7D nr. 1.4

$f(x, y) = \sin(x^2y) - x \cos(y^2)$ in het punt $(0, 0)$, graad 3.

eerste orde:

$$f(0, 0) = \sin(0) - 0 \cdot \cos(0) = 0$$

$$\frac{\partial f}{\partial x} = \cos(x^2y) \cdot 2xy - \cos(y^2) \implies \frac{\partial f}{\partial x}(0, 0) = 0 - 1 = -1$$

$$\frac{\partial f}{\partial y} = \cos(x^2y) \cdot x^2 + x \sin(y^2) \cdot 2y \implies \frac{\partial f}{\partial y}(0, 0) = 0 + 0 = 0$$

tweede orde:

$$\frac{\partial^2 f}{\partial x^2} = 2y \cos(x^2y) - 4x^2y^2 \sin(x^2y) \implies \frac{\partial^2 f}{\partial x^2}(0, 0) = 0$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2x \cos(x^2y) - 2x^3y \sin(x^2y) + 2y \sin(y^2) \implies \frac{\partial^2 f}{\partial x \partial y}(0, 0) = 0$$

$$\frac{\partial^2 f}{\partial y^2} = -x^4 \sin(x^2y) + 2x \sin(y^2) + 4xy^2 \cos(y^2) \implies \frac{\partial^2 f}{\partial y^2}(0, 0) = 0$$

derde orde:

$$\frac{\partial^3 f}{\partial x^3} = -4xy^2 \sin(x^2y) - 8xy^2 \sin(x^2y) - 8x^3y^3 \cos(x^2y) \implies \frac{\partial^3 f}{\partial x^3}(0, 0) = 0$$

$$\frac{\partial^3 f}{\partial x^2 \partial y} = 2 \cos(x^2y) - 2x^2y \sin(x^2y) - 8x^2y \sin(x^2y) - 4x^4y^2 \cos(x^2y) \implies \frac{\partial^3 f}{\partial x^2 \partial y}(0, 0) = 2$$

$$\frac{\partial^3 f}{\partial x \partial y^2} = -4x^3 \sin(x^2y) - 2x^5y \cos(x^2y) + 2 \sin(y^2) + 4y^2 \cos(y^2) \implies \frac{\partial^3 f}{\partial x \partial y^2}(0, 0) = 0$$

$$\frac{\partial^3 f}{\partial y^3} = -x^6 \cos(x^2y) + 4xy \cos(y^2) + 8xy \cos(y^2) - 8xy^3 \sin(y^2) \implies \frac{\partial^3 f}{\partial y^3}(0, 0) = 0$$

Taylorpolynoom:

$$\begin{aligned} P_3(x, y) &= f(0, 0) + \left(\frac{\partial f}{\partial x}x + \frac{\partial f}{\partial y}y \right) + \frac{1}{2} \left(\frac{\partial^2 f}{\partial x^2}x^2 + 2 \frac{\partial^2 f}{\partial x \partial y}xy + \frac{\partial^2 f}{\partial y^2}y^2 \right) + \\ &\frac{1}{6} \left(\frac{\partial^3 f}{\partial x^3}x^3 + 3 \frac{\partial^3 f}{\partial x^2 \partial y}x^2y + 3 \frac{\partial^3 f}{\partial x \partial y^2}xy^2 + \frac{\partial^3 f}{\partial y^3}y^3 \right) \\ P_3(x, y) &= 0 + (-1 \cdot x + 0 \cdot y) + 0 + \frac{1}{6}(0 + 3 \cdot 2 \cdot x^2y + 0 + 0) \end{aligned}$$

$$P_3(x, y) = -x + x^2y$$

References